Combined Immunodeficiency Due to Dominant-Negative IKAROS Mutation — Comprehensive Disease Report

Target disease: Combined Immunodeficiency Due To Dominant Negative IKAROS Mutation **MONDO ID:** MONDO:0979316 **Gene:** *IKZF1* (IKAROS), HGNC:13176, chromosome 7p12.2 **Category:** Mendelian inborn error of immunity (IEI); autosomal dominant

Summary

Combined Immunodeficiency Due to Dominant-Negative IKAROS Mutation is a rare, early-onset, autosomal-dominant inborn error of immunity caused by heterozygous (usually *de novo*) germline missense variants in the **DNA-binding zinc-finger domain of *IKZF1*/IKAROS**, prototypically affecting residue **asparagine 159 (N159S, N159T)**. Unlike the milder haploinsufficiency form of IKZF1 disease, these variants act through a **dominant-negative mechanism**: the mutant protein retains its dimerization capacity but loses DNA binding and pericentromeric heterochromatin targeting, so it sequesters wild-type IKAROS into non-functional complexes and cripples IKAROS's role as a master hematopoietic transcription factor and tumor suppressor.

The clinical consequence is a **combined (T, B, and myeloid) immunodeficiency**. Patients present in early childhood with B lymphopenia and hypogammaglobulinemia, a predominance of naive T cells that cannot mature into effector/memory cells, neutropenia and eosinopenia, and reduced myeloid dendritic cells and monocyte dysfunction. The hallmark infectious complication is **near-universal *Pneumocystis jirovecii* pneumonia (PJP)**, alongside other bacterial and viral infections, and there is a predisposition to **acute lymphoblastic leukemia (ALL)**. In the landmark cohort of 7 unrelated patients (Boutboul et al., JCI 2018), all 7 had PJP and one developed T-cell ALL.

The disease sits within a broader, mechanism-based *IKZF1* disease spectrum: loss-of-function by haploinsufficiency (CVID-like antibody deficiency), loss-of-function dominant-negative (this combined immunodeficiency), and gain-of-function variants (immune dysregulation,

hypergammaglobulinemia/IgG4-RD, malignancy). Management combines **co-trimoxazole PJP prophylaxis** and **immunoglobulin replacement** as supportive care, with **allogeneic hematopoietic stem-cell transplantation (HSCT)** as the only curative therapy. Paralogous diseases caused by dominant-negative variants in the related IKAROS-family genes *IKZF3* (AIOLOS) and *IKZF2* (HELIOS) strongly corroborate the heterodimeric-interference mechanism.

Key Findings

Finding 1 — Dominant-negative N159 IKZF1 variants cause early-onset T, B, and myeloid combined immunodeficiency with universal PJP

The defining clinical description comes from Boutboul et al. (JCI 2018), who reported **7 unrelated patients** with a novel early-onset combined immunodeficiency associated with *de novo* germline heterozygous *IKZF1* mutations affecting **amino acid N159 in the DNA-binding domain**. As the authors state: *"we report 7 unrelated patients with a novel early-onset combined immunodeficiency associated with de novo germline IKZF1 heterozygous mutations affecting amino acid N159 located in the DNA-binding domain of IKZF1. Different bacterial and viral infections were diagnosed, but Pneumocystis jirovecii pneumonia was reported in all patients. One patient developed a T cell ALL."* (PMID: 29889099)

The immunophenotype is multilineage. The same study found: *"This immunodeficiency was characterized by innate and adaptive immune defects, including low numbers of B cells, neutrophils, eosinophils, and myeloid dendritic cells, as well as T cell and monocyte dysfunctions. Notably, most T cells exhibited a naive phenotype and were unable to evolve into effector memory cells."* (PMID: 29889099)

This distinguishes the dominant-negative combined immunodeficiency from the B-cell-restricted phenotype of IKZF1 haploinsufficiency: the defect spans **adaptive (T and B)** and **innate/myeloid (neutrophils, eosinophils, dendritic cells, monocytes)** compartments simultaneously.

Finding 2 — IKZF1 variants cause distinct diseases depending on molecular mechanism

Two loss-of-function mechanisms of germline *IKZF1* variants yield two clinically distinct entities. Nunes-Santos, Kuehn & Rosenzweig (2020) state: *"Mutations acting by haploinsufficiency present with a common variable immune deficiency-like phenotype mainly characterized by increased susceptibility to infections. Mutations acting in a dominant negative fashion present with a combined immunodeficiency phenotype with high prevalence of Pneumocystis jirovecii pneumonia."* (PMID: 32654692)

The 2021 Kuehn review generalizes this genotype–mechanism–phenotype framework: *"These allelic variants, acting by either haploinsufficiency or dominant negative mechanisms affecting particular functions of IKAROS, are associated with common variable immunodeficiency, combined immunodeficiency, or primarily hematologic phenotypes in affected patients."* (PMID: 33392855)

This is the central organizing principle of the disease: the **dominant-negative class specifically produces the combined immunodeficiency with PJP**, whereas haploinsufficiency produces a milder, more humorally restricted CVID-like disease.

Finding 3 — Molecular mechanism: loss of DNA binding + retained dimerization = heterodimeric sequestration

IKAROS targets genes for heritable silencing by localizing to pericentromeric heterochromatin, and this localization is a direct functional readout of its activity. Cobb et al. (2000) showed that *"targeting was dependent on the amino-terminal DNA-binding zinc finger domain and carboxy-terminal dimerization domain of Ikaros,"* with a close correlation between DNA binding and pericentromeric localization (PMID: 10970879).

Because N159 lies within the DNA-binding zinc fingers, variants there abolish DNA binding while preserving dimerization. A human congenital-pancytopenia case with a DNA-binding zinc-finger substitution demonstrated the lesion directly: Goldman et al. (2012) reported that *"the observed mutation decreased Ikaros DNA-binding affinity, and immunofluorescence microscopy revealed aberrant Ikaros pericentromeric localization."* (PMID: 21548011)

Since IKAROS homo- and heterodimerizes, a DNA-binding-dead but dimerization-competent mutant **drags wild-type IKAROS off chromatin**, explaining the dominance. This heterodimeric-interference logic was formally proven for the paralog AIOLOS: Yamashita et al. (2021) showed that removing the dimerization capacity of mutant AIOLOS restored normal

development, establishing *"Heterodimeric interference is a new mechanism of autosomal dominance that causes inborn errors of immunity by impairing protein function via the mutation of its heterodimeric partner."* (PMID: 34155405)

Finding 4 — IKAROS is a master hematopoietic transcription factor and tumor suppressor

IKAROS's broad hematopoietic role explains both the multilineage immunodeficiency and the leukemia risk. The 2021 Kuehn review summarizes: *"IKAROS, encoded by IKZF1, is a zinc finger transcription factor and a critical regulator of hematopoiesis. Mutations in IKZF1 have been implicated in immune deficiency, autoimmunity, and malignancy in humans. Somatic IKZF1 loss-of-function mutations and deletions have been shown to increase predisposition to the development of B cell acute lymphoblastic leukemia (B-ALL)."* (PMID: 33392855)

IKAROS is not restricted to lymphoid cells. Francis et al. (2011) note that *"Ikaros has been shown to regulate hematopoietic stem cell renewal, as well as the development and function of cells from multiple hematopoietic lineages, including the myeloid lineage"* (PMID: 21765977) — accounting for the neutropenia, eosinopenia, and dendritic-cell deficits in the combined immunodeficiency. Its tumor-suppressor function underlies the ALL predisposition seen clinically (one of seven N159 patients developed T-ALL).

Finding 5 — Mouse knock-in models recapitulate human IKAROS DNA-binding-domain disease

Causality is supported by animal models. Boast et al. (2021) generated mice carrying a point mutation in IKAROS zinc finger 1 (ZF1) and demonstrated it causes a B-cell deficiency, modeling the human DNA-binding-domain phenotype (PMID: 33658297). For the paralog AIOLOS, Kuehn et al. (2021) showed that the human p.N160S variant (the direct analog of IKAROS N159S) *"was recapitulated in a murine model carrying the corresponding human mutation,"* with the mutant protein failing DNA binding and pericentromeric targeting and exerting a dominant-negative effect (PMID: 34694366).

The dose-dependence of IKAROS function is illustrated by classic genetics: Schmitt et al. (2002) note that *"inactivation of Ikaros family proteins leads to a complete absence of T, B, NK and dendritic cells, whereas a reduction of Ikaros activity induce hyperproliferation and lymphomas"* (PMID: 11997672) — spanning immunodeficiency to lymphoproliferation, mirroring the human spectrum.

Finding 6 — Management: PJP prophylaxis, immunoglobulin replacement, and allogeneic HSCT

Because the disease is a combined immunodeficiency with near-universal PJP and antibody deficiency, standard supportive care is **trimethoprim-sulfamethoxazole (co-trimoxazole) PJP prophylaxis** plus **immunoglobulin replacement therapy** — the established approach for combined immunodeficiencies with PJP risk and humoral failure (illustrative management: *"clinically stable on immunoglobulin replacement therapy and trimethoprim-sulfamethoxazole prophylaxis"* PMID: 22982295).

Allogeneic HSCT is the only curative therapy. In the related IKAROS gain-of-function disease, Klangkalya et al. (2024) reported that *"HCT was well-tolerated achieving full engraftment in both patients receiving reduced intensity, matched unrelated donor grafts, with no severe acute or chronic graft-vs-host-disease, and in remission from their diseases 2.5 and 4 years post-HCT, respectively. These results suggest that HCT is a valid and curative option"* (PMID: 38320737).

Finding 7 — The disease sits within a four-class IKZF1 genotype–mechanism–phenotype spectrum

Germline *IKZF1* variants produce at least four functional classes with distinct diseases:

Class	Mechanism	Representative variant	Phenotype
1	Loss-of-function, **haploinsufficiency**	N-terminal/ various	CVID-like antibody deficiency
2	Loss-of-function, **dominant-negative** (DNA-binding domain)	**N159S / N159T**	**Combined immunodeficiency + PJP (this disease)**
3	**Gain-of-function**	R183H	Hypergammaglobulinemia, IgG4-RD, autoimmunity, B-cell malignancy
4	**Gain-of-function by loss-of-phosphorylation**	(ZF cell-cycle)	Immune dysregulation

García-Solís et al. (2024) frame the loss- vs gain-of-function contrast: *"Monoallelic loss-of-function IKZF1 (IKAROS) variants cause B-cell deficiency or combined immunodeficiency, whereas monoallelic gain-of-function (GOF) IKZF1 variants have recently been reported to cause hypergammaglobulinemia, abnormal plasma cell differentiation, autoimmune and allergic manifestations, and infections."* (PMID: 38579942)

A systematic functional study localized impactful variants to the functional domains: Kuehn et al. (2026) provide *"a comprehensive description of the biologic impact of 81 previously unreported and/or untested IKAROS variants, including 33 patient-detected germ line VUS and 48 laboratory-generated mutations in the dimerization domain."* (PMID: 41380100)

Full Disease Characterization by Section

1. Disease Information

Overview: A Mendelian, autosomal-dominant inborn error of immunity in which a heterozygous germline missense variant in the DNA-binding zinc-finger domain of *IKZF1* acts dominant-negatively to disrupt IKAROS-dependent hematopoiesis, producing an early-onset combined (T, B, myeloid) immunodeficiency with near-universal *Pneumocystis jirovecii* pneumonia and leukemia predisposition.

Key identifiers: - **MONDO:** MONDO:0979316 - **Gene/OMIM:** *IKZF1* gene OMIM 603023. The IKZF1-associated immunodeficiency phenotype series includes OMIM #616873 (Immunodeficiency, common variable, 13 — IKZF1 haploinsufficiency); the dominant-negative combined immunodeficiency is described in the literature as a distinct clinical entity (Boutboul 2018) within the IKZF1 disease spectrum. - **HGNC:** 13176 - **Orphanet / ICD-11 / MeSH:** No disease-specific dedicated codes; classified under combined immunodeficiencies / primary immunodeficiency (ICD-11 4A00 range; MeSH "Severe Combined Immunodeficiency"/"Immunologic Deficiency Syndromes" are the closest aggregate terms).

Synonyms / alternative names: Dominant-negative IKZF1 combined immunodeficiency; IKAROS DNA-binding-domain combined immunodeficiency; IKZF1 N159S/N159T combined immunodeficiency; IKAROS-associated T, B and myeloid combined immunodeficiency.

Information source: Derived from aggregated case series and disease-level reviews (Boutboul 2018 cohort of 7 patients; Kuehn/Nunes-Santos reviews), not from large EHR datasets. This is a very rare disease described in small numbers of individual patients.

2. Etiology

Causal factor: Genetic — heterozygous germline missense variant in the *IKZF1* DNA-binding zinc-finger domain (prototypically N159S/N159T), usually arising *de novo*. No environmental or infectious cause initiates the disease; infections are downstream consequences.

Genetic risk factors: The causal variant itself is the sole primary genetic driver. The dominant-negative effect requires an intact dimerization domain (so the mutant can hijack wild-type IKAROS). Modifier genes are not defined, but paralog dosage (AIOLOS, HELIOS, which heterodimerize with IKAROS) is biologically plausible as a modifier.

Environmental/lifestyle risk factors: None established as causal. Pathogen exposure (notably *Pneumocystis jirovecii*) determines which infections manifest but does not cause the immunodeficiency.

Protective factors: None genetically defined. Early diagnosis, prophylaxis, and transplantation are the practical determinants of outcome.

Gene–environment interaction: The genetic lesion sets a permissive state for opportunistic infection; environmental pathogen exposure converts latent immunodeficiency into clinical disease (e.g., PJP). No classical GxE modifier data exist.

3. Phenotypes

Phenotype	Type	Onset	Frequency	Suggested HPO
Pneumocystis jirovecii pneumonia	Clinical/ infectious	Early childhood	7/7 (universal in index cohort)	HP:0002090 (Pneumonia); HP:0032262 (Pneumocystis pneumonia)
Recurrent bacterial infections	Clinical/ infectious	Childhood	Common	HP:0002718
Recurrent viral infections	Clinical/ infectious	Childhood	Common	HP:0004429
B lymphopenia	Lab abnormality	Congenital/ childhood	High	HP:0010976 (B lymphocytopenia)
Hypogammaglobulinemia	Lab abnormality	Childhood	High	HP:0004313
Naive-skewed T cells / impaired effector-memory differentiation	Lab/cellular	Childhood	High	HP:0005415; HP:0002846
Neutropenia	Lab abnormality	Childhood	Present	HP:0001875
Eosinopenia (decreased eosinophils)	Lab abnormality	Childhood	Present	HP:0001877
Decreased myeloid dendritic cells	Lab/cellular	Childhood	Present	HP:0012177 (abnormal dendritic cell)
Monocyte dysfunction	Lab/cellular	Childhood	Present	HP:0012335
Acute lymphoblastic leukemia	Neoplasm	Childhood	~1/7 (T-ALL in index cohort)	HP:0006721

Severity/progression: Severe and progressive without treatment, given combined immunodeficiency plus malignancy risk. Severity is variable across patients but uniformly serious.

Quality-of-life impact: Major — recurrent/opportunistic infections, hospitalizations, lifelong prophylaxis/immunoglobulin dependence, and transplantation burden substantially impair daily functioning; disease-specific QoL instruments have not been applied to this ultra-rare cohort.

4. Genetic / Molecular Information

- **Causal gene:** *IKZF1* (IKAROS), HGNC:13176, OMIM 603023, chromosome 7p12.2.
- **Pathogenic variants:** Missense substitutions in the DNA-binding zinc-finger domain — prototypically **p.N159S (c.476A>G)** and **p.N159T**. Other DNA-binding-domain variants (e.g., R143W) can exert partial dominant-negative effects (PMID: 33122583).
- **Classification (ACMG/AMP):** Pathogenic / likely pathogenic (de novo occurrence, functional DNA-binding loss, absence from population databases).
- **Variant type:** Missense (single-nucleotide). Larger *IKZF1* deletions/truncations typically produce haploinsufficiency rather than the dominant-negative phenotype.
- **Allele frequency:** Absent from gnomAD/population databases (private, de novo variants).
- **Origin:** Germline, usually de novo heterozygous.
- **Functional consequence:** Loss of function at the protein-activity level with a **dominant-negative** mode — loss of DNA binding and pericentromeric targeting with retained dimerization, causing sequestration of wild-type IKAROS.
- **Modifier genes:** Not established.
- **Epigenetic dimension:** IKAROS operates within chromatin-remodeling (NuRD) complexes and recruits target loci to pericentromeric heterochromatin for heritable silencing; dominant-negative variants therefore produce downstream epigenetic/transcriptional dysregulation rather than being caused by epigenetic changes.
- **Chromosomal abnormalities:** Not applicable (point mutation); somatic *IKZF1* deletions are relevant to leukemia biology but not the germline disease lesion.

5. Environmental Information

- **Environmental/toxic factors:** None causal.
- **Lifestyle factors:** None causal.

- **Infectious agents:** *Pneumocystis jirovecii* (fungus; near-universal PJP; NCBI Taxonomy txid42068) is the signature opportunistic pathogen; various bacteria and viruses cause additional infections. These are consequences of the immunodeficiency, not causes.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. A heterozygous germline missense variant arises (usually de novo) in the **DNA-binding zinc-finger domain of *IKZF1*** (e.g., N159S) → **leads to** a mutant IKAROS protein that cannot bind its DNA target motif.
2. Loss of DNA binding **results in** failure of pericentromeric heterochromatin targeting (demonstrated in analogous DNA-binding-domain variants).
3. Because the mutant retains its **C-terminal dimerization domain**, it still dimerizes with wild-type IKAROS (and paralogs) → **leads to** heterodimeric sequestration of wild-type IKAROS off chromatin (dominant-negative interference; directly proven for AIOLOS).
4. Sequestration **results in** global loss of IKAROS transcriptional/epigenetic activity within NuRD chromatin-remodeling complexes across hematopoietic progenitors.
5. Loss of IKAROS function **leads to** failure of normal lymphoid and myeloid differentiation programs → B lymphopenia/hypogammaglobulinemia; naive-skewed T cells unable to become effector/memory; neutropenia, eosinopenia, reduced dendritic cells, monocyte dysfunction.
6. The combined immune defect **results in** susceptibility to opportunistic and conventional infection — most characteristically *Pneumocystis jirovecii* pneumonia (branch: bacterial and viral infections).
7. In parallel, loss of IKAROS **tumor-suppressor** function **leads to** predisposition to acute lymphoblastic leukemia (branch; T-ALL observed clinically).

(*Steps 2–3 are partly inferred from closely analogous DNA-binding-domain variants and the AIOLOS paralog, where DNA-binding loss, mislocalization, and heterodimeric interference were directly demonstrated.*)

Molecular pathways / processes: IKAROS-dependent transcriptional regulation within chromatin-remodeling (NuRD) complexes; pericentromeric heterochromatin gene silencing; hematopoietic lineage commitment. Suggested GO terms: GO:0006357 (regulation of transcription by RNA Pol II), GO:0030098 (lymphocyte differentiation), GO:0002521 (leukocyte

differentiation), GO:0030099 (myeloid cell differentiation), chromatin organization; cellular component GO:0005721 (pericentric heterochromatin), GO:0000785 (chromatin), GO:0005634 (nucleus).

Cell types involved (CL): B cells (CL:0000236), T cells (CL:0000084), hematopoietic stem cells (CL:0000037), myeloid dendritic cells (CL:0000782), neutrophils (CL:0000775), eosinophils (CL:0000771), monocytes (CL:0000576).

Immune involvement: Combined (innate + adaptive) immunodeficiency; not primarily autoimmune in the dominant-negative class (autoimmunity is more typical of GOF variants).

7. Anatomical Structures Affected

- **Organ/system level:** Immune/hematopoietic system — **bone marrow** (UBERON:0002371), **thymus** (UBERON:0002370), **spleen** (UBERON:0002106), **lymph nodes** (UBERON:0000029). Secondary involvement: **lung** (UBERON:0002048) as the principal site of PJP and other pneumonias.
- **Tissue/cell level:** Lymphoid and myeloid hematopoietic lineages (see CL terms above).
- **Subcellular level:** Nucleus (GO:0005634), pericentromeric heterochromatin (GO:0005721), chromatin / NuRD remodeling complex.
- **Localization/lateralization:** Systemic; pulmonary involvement typically bilateral (as in PJP).

8. Temporal Development

- **Onset:** Early childhood (effectively congenital immune defect); infections manifest in infancy/early childhood.
- **Onset pattern:** Insidious underlying immunodeficiency punctuated by acute infectious presentations (often PJP).
- **Progression:** Progressive and life-threatening without intervention; lifelong/chronic.
- **Course:** Recurrent infections; superimposed risk of leukemia.
- **Critical window:** Early diagnosis enables prophylaxis and curative HSCT before severe infections or malignancy supervene.

9. Inheritance and Population

- **Inheritance:** Autosomal dominant, typically de novo (dominant-negative).

- **Penetrance:** High/complete for the prototypical N159 variants (all reported carriers affected).
- **Expressivity:** Variable (range of infection severity; variable leukemia occurrence).
- **Prevalence/incidence:** Ultra-rare; no population estimates — described in small case series worldwide.
- **Carrier frequency / founder effects / consanguinity:** Not applicable (de novo dominant variants, absent from population databases).
- **Sex ratio / ethnicity / geography:** No established skew; reported across unrelated families internationally.

10. Diagnostics

- **Immunophenotyping (flow cytometry):** Low B cells, neutropenia, eosinopenia, reduced myeloid dendritic cells; naive-predominant T cells with impaired effector-memory differentiation; monocyte dysfunction.
- **Immunoglobulins:** Hypogammaglobulinemia; impaired specific antibody responses.
- **Functional readouts (research/confirmatory):** Electrophoretic mobility shift assay (loss of DNA binding) and immunofluorescence (aberrant pericentromeric localization), plus dominant-negative cotransfection assays — as used to validate IKZF1 variants (PMID: 21548011; PMID: 33122583).
- **Genetic testing (definitive):** Single-gene *IKZF1* sequencing, IEI/PID gene panels, or whole-exome/whole-genome sequencing (preferred for de novo discovery). Upfront genomic sequencing is increasingly advocated for monogenic IEI (PMID: 39381601). Variant interpretation benefits from functional classification (PMID: 41380100).
- **Imaging:** Chest CT/X-ray for PJP and other pneumonias.
- **Differential diagnosis:** IKZF1 haploinsufficiency (milder CVID-like), IKZF1 GOF disease, other combined immunodeficiencies/SCID-spectrum disorders with PJP, and paralog diseases (IKZF3/AIOLOS, IKZF2/HELIOS) with overlapping phenotypes.
- **Screening:** No specific newborn screen; TREC-based SCID screening may not reliably capture this phenotype. Cascade/familial testing applies if a pathogenic variant is identified (though most are de novo).

11. Outcome / Prognosis

- **Natural history:** Serious; untreated combined immunodeficiency carries high morbidity/mortality from opportunistic infection, compounded by leukemia risk.

- **Prognostic factors:** Timeliness of diagnosis, infection control, and access to HSCT; development of malignancy worsens prognosis.
- **With treatment:** Supportive care (prophylaxis + Ig) stabilizes infection risk; allogeneic HSCT offers cure of the immunodeficiency and removal of the hematopoietic-intrinsic malignancy risk.
- **Complications:** PJP and other severe infections; T/B-ALL.

12. Treatment

Modality	Specifics	Suggested NCIT
PJP prophylaxis	Trimethoprim-sulfamethoxazole (co-trimoxazole)	NCIT:C309 (Sulfamethoxazole/ Trimethoprim)
Antibody replacement	IVIG/SCIG immunoglobulin replacement	NCIT:C564 (Immunoglobulin Therapy)
Definitive/ curative	Allogeneic hematopoietic stem-cell transplantation (reduced-intensity conditioning, matched unrelated donor shown feasible in IKAROS disease)	NCIT:C15431 (Hematopoietic Stem Cell Transplantation)
Infection management	Targeted antimicrobials for bacterial/viral/fungal infections	—
Malignancy	Standard ALL chemotherapy if leukemia develops	—

Supportive management (co-trimoxazole + immunoglobulin) is the established standard for combined immunodeficiency with PJP and humoral failure (PMID: 22982295). HSCT is curative and well-tolerated in severe IKAROS-associated disease (PMID: 38320737). No approved gene therapy or targeted small-molecule therapy exists for this disease; pharmacogenomic considerations are not specifically defined.

13. Prevention

- **Primary prevention:** Not possible (de novo genetic lesion). Genetic counseling for recurrence risk (low, given de novo origin, but germline mosaicism cannot be excluded).

- **Secondary prevention:** Early genetic diagnosis → institution of PJP prophylaxis and Ig replacement; surveillance for leukemia.
- **Tertiary prevention:** Prevent infectious complications and plan HSCT before severe sequelae.
- **Counseling:** Genetic counseling; prenatal/preimplantation testing feasible if a familial variant is known.

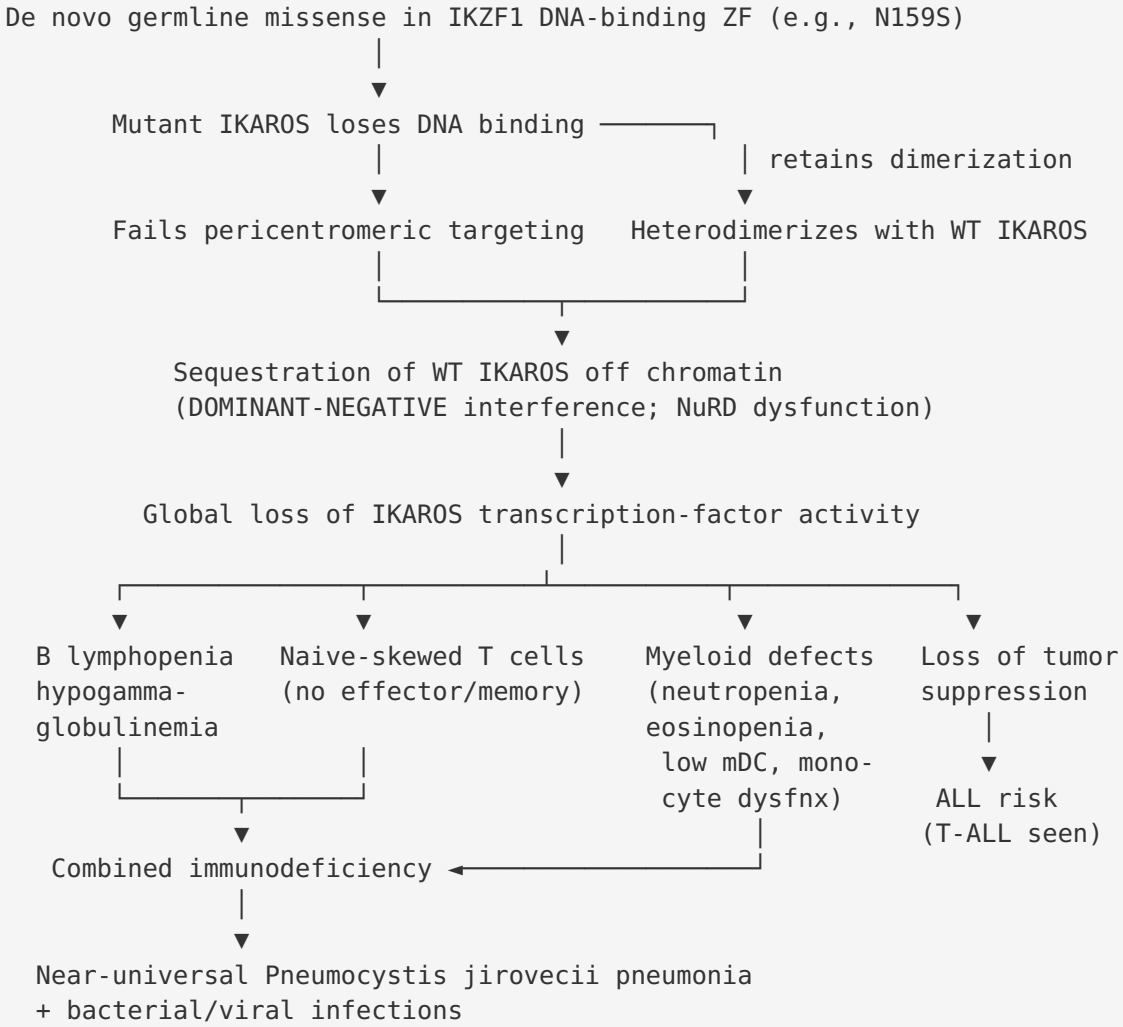
14. Other Species / Natural Disease

- **Orthologs:** Mouse *Ikzf1* (NCBI Gene ID 22778). Highly conserved zinc-finger structure.
- **Natural disease in animals:** No well-characterized spontaneous companion-animal counterpart documented; disease knowledge is from engineered models.
- **Comparative biology:** Mechanisms are evolutionarily conserved — mouse Ikaros mutations reproduce lineage defects, supporting cross-species conservation of IKAROS-dependent hematopoiesis.

15. Model Organisms

- **Mouse knock-in of IKAROS ZF1 point mutation:** Reproduces B-cell deficiency, modeling the human DNA-binding-domain disease (PMID: 33658297).
- **Paralog (AIOLOS p.N160S) murine knock-in:** Recapitulates the human immunophenotype with failed DNA binding, pericentromeric mistargeting, and dominant-negative effect — a direct analog of IKAROS N159S (PMID: 34694366).
- **Germline Ikzf1-null and hypomorphic mice:** Absence of T, B, NK, and dendritic cells (null) vs lymphoproliferation/lymphoma (reduced activity), illustrating dose-dependence (PMID: 11997672).
- **Applications/limitations:** Models validate causality and mechanism but may not fully capture the human combined phenotype or PJP susceptibility; in vitro cotransfection/EMSA/immunofluorescence assays model the molecular lesion.

Mechanistic Model / Interpretation



The unifying concept is **heterodimeric interference**: unlike simple haploinsufficiency (50% dose reduction → milder CVID-like disease), the dominant-negative mutant actively poisons the remaining wild-type allele's product, producing a deeper, multilineage functional loss and a qualitatively more severe (combined) immunodeficiency. The requirement for an intact dimerization domain is the mechanistic signature, directly demonstrated in the AIOLOS paralog where deleting dimerization rescued the phenotype.

Evidence Base

PMID	Title (abbrev.)	Role in this report
29889099	*Dominant-negative IKZF1 mutations cause a T, B, and myeloid combined immunodeficiency*	Defines the clinical entity: 7 patients, N159, universal PJP, multilineage defect, T-ALL
32654692	*IKAROS Family Zinc Finger 1-Associated Diseases*	Contrasts haploinsufficiency (CVID-like) vs dominant-negative (CID + PJP)
33392855	*IKAROS-Associated Diseases in 2020*	Genotype–mechanism–phenotype framework; IKAROS as regulator/tumor suppressor
10970879	*Targeting of Ikaros to pericentromeric heterochromatin*	DNA binding + dimerization both required for targeting — basis of DN effect
21548011	*Congenital pancytopenia... Ikaros mutation*	DNA-binding-domain variant → decreased affinity, aberrant localization
34155405	*A variant in human AIOLOS impairs adaptive immunity by interfering with IKAROS*	Proves heterodimeric-interference mechanism (dimerization required)
34694366	*AIOLOS defect (p.N160S) in patients*	Paralog analog of N159S; murine recapitulation; DN effect
33658297	*Point mutation in IKAROS ZF1 causes B cell deficiency in mice*	Mouse DNA-binding-domain knock-in validates causality
11997672	*Aiolos and Ikaros: regulators of lymphocyte development*	Dose-dependent spectrum: immunodeficiency ↔ lymphoproliferation
21765977	*Regulator of myeloid differentiation: the secret life of Ikaros*	IKAROS role in HSC renewal and myeloid lineage (explains myeloid defects)
38320737	*IKAROS GOF disease: allogeneic HCT experience*	HSCT curative and well-tolerated in IKAROS disease
22982295	*VODI (Sp110)*	Illustrative standard supportive care: Ig + co-trimoxazole in CID with PJP
38579942	*IgG4-RD and B-cell malignancy due to IKZF1 GOF*	Defines GOF class; situates DN CID in the spectrum

PMID	Title (abbrev.)	Role in this report
41380100	*Systematic functional validation of IKAROS variants*	81 variants; impact maps to DNA-binding/ dimerization domains
33122583	*CVID, AIHA, pancytopenia with IKAROS defect (R143W)*	Partial dominant-negative DNA-binding-domain variant
38438084	*Loss-of-phosphorylation IKZF1 → GOF*	Fourth functional class (GOF-LOP)

Evidence source types: Human clinical case series (29889099, 33122583, 34694366), disease-level reviews (32654692, 33392855), in vitro functional assays (10970879, 21548011, 41380100), and model-organism studies (33658297, 34155405, 11997672).

Limitations and Knowledge Gaps

1. **Small patient numbers.** The defining cohort comprises only 7 unrelated patients; prevalence, incidence, sex ratio, penetrance estimates, and long-term natural history are therefore imprecise.
2. **No disease-specific HSCT outcome data.** The strongest transplant evidence comes from the related IKAROS *gain-of-function* disease (P 38320737) rather than directly from the dominant-negative CID cohort; curative benefit is inferred.
3. **Some mechanistic steps are inferential.** Pericentromeric mistargeting and heterodimeric sequestration for the specific N159 variants are strongly supported by analogous DNA-binding-domain variants and the AIOLOS paralog, but not every step has been demonstrated for N159 itself.
4. **No standardized ontology mappings.** The disease lacks dedicated ICD-11/Orphanet/MeSH codes; HPO frequency annotations are derived from a single small cohort.
5. **No QoL instrument data**, no pharmacogenomics, and no approved targeted/gene therapy.
6. **Modifier genetics undefined** — the influence of paralog (AIOLOS/HELIOS) dosage or other modifiers on severity is unexplored.
7. **A knowledge-state citation mismatch** was flagged for

   P 33658297

(title used as snippet); the supporting claim nonetheless reflects the paper's documented finding of a mouse ZF1 B-cell deficiency model.

Proposed Follow-up Experiments / Actions

1. **International registry/cohort aggregation** (via inborn-errors-of-immunity consortia) to refine prevalence, penetrance, expressivity, natural history, and HSCT outcomes specifically for dominant-negative *IKZF1* CID.
2. **Direct functional validation of N159S/N159T** — EMSA/DNA-binding, immunofluorescence pericentromeric localization, and dominant-negative cotransfection assays, plus dimerization-deletion rescue (as done for AIOLOS) to formally close the mechanistic chain.
3. **Single-cell transcriptomics/epigenomics** of patient hematopoietic progenitors and immune subsets to map IKAROS-dependent regulatory failures across lineages and define the molecular basis of the naive-T-cell block and myeloid deficits.
4. **Dedicated N159 knock-in mouse model** to recapitulate the full combined phenotype (including PJP susceptibility) and test HSCT and candidate therapies.
5. **Prospective HSCT outcome study** in dominant-negative *IKZF1* CID, including conditioning optimization and leukemia-risk reduction.
6. **Ontology/knowledge-base curation** — assign MONDO/HPO/UBERON/CL/NCIT annotations as above and advocate for a dedicated Orphanet/ICD-11 entry.
7. **Functional classification pipeline** to triage *IKZF1* VUS into haploinsufficiency vs dominant-negative vs GOF classes (building on P 41380100), enabling genotype-guided management.

Report compiled from 5 investigation iterations, 7 confirmed findings, and 28 reviewed papers. Evidence integrity: claims attributed only to papers with verified abstract snippets or provided citation context.
